MIL-D-19151A(OS)
AMENDMENT 2
20 June 1985
SUPERSEDING
AMENDMENT 1
6 July 1972

DETONATOR MARK 57 MODS 1 AND 2

This amendment forms a part of Military Specification MIL-D-19151A(OS), dated 10 April 1972, and is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

Page 2

Para 2.1, after MIL-STD-331, add:

"MIL-STD-202 Test Methods for Electronic and Electrical Component Parts."

Page 7

Para 3.2.5.5.1, line 4, change "0.023" to "0.021" and change "0.030" to "0.029".

line 6, change "0.021" to "0.019" and change "0.028" to "0.027".

After para 3.2.5.8, add:

"3.2.5.9 Radiographic Inspection. The detonator shall meet the following requirements when radiographically inspected. The detonator shall be tested in accordance with 4.5.12.

a. The pentaerythritol tetranitrate (PETN) charge shall be in accordance with Dwg 1664933 or 2508061 as applicable and shall be identifiable and without cracks or voids.

b. The lead azide ignition increments shall be in accordance with Dwg 1664935 or 2508062 as applicable and shall be identifiable and without cracks or voids.

c. The charge holder Dwg 959567 shall be firmly seated against the bridgewire assembly Dwg 1409497 or 2508064 as applicable.

d. The lead azide ignition charge shall be in contact with the lead styphnate flash charge (see Dwg 1664935 or 2508062 as applicable).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640-5000.

FSC 1356

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e. The ignition assembly Dwg 1664935 or 2508062 as applicable shall be seated flush against the PETN charge in the cup assembly Dwg 1664933 or 2508061 as applicable. (.005 inch maximum gap allowable between PETN charge and lead azide charge).

f. There shall be no evidence of metal slivers, wire clippings or other foreign material within the detonator assembly.

g. There shall be no evidence of malformed parts within the detonator assembly."

Page 12

Para 4.5.6, second line, Change "(see 6.2.3)" to "(see 6.3.1)".

Page 14

After para 4.5.11, add:

"4.5.12 Radiographic Test (M112). The detonators shall be tested in accordance with MIL-STD-202, Method 209 in the directions indicated on Dwg 1261834 or 2508060 as applicable. Any detonator which fails to meet all the requirements of 3.2.5.9 shall be classified as defective. Acceptance shall be in accordance with Figure 1 through Figure 4 as applicable."

Page 15

Change paragraph "6.1" to "6.2" and add:

"f. Mod of detonator (1.1)

g. Special markings required (5.4)

h. That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable."

Add new paragraph 6.1 as follows:

"6.1 Intended Use. Detonators Mark 57 Mods 1 and 2 are used in Arming Device Mark 2 Mods 1, 3, and 5."

Page 16

Delete paragraph 6.3 in its entirety and change section "6.2" to "6.3".

Page 18

Change "*M102	4.5.2"
to "*M102(V)	4.5.2"
Change M107 (0.25% AQL)	4.5.7"
to M107(V)	4.5.7"
Change M108(V) (0.25% AQL)	4.5.8.1"
to M108(V)	4.5.8.1"
After "*M111(V)	3.2.4"
Add "M112 (sampling and acceptance in accordance with figure 1, 2, 3, or 4)"	4.5.12

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After "*M111 3.2.4"
Add "M112 (sampling and acceptance in accordance with Figure 1,2,3 or 4)" 4.5.12

Page 22

Figure 1, after "Dimensional and Material Inspection" block add:

"
Radiographic Inspection 4.5.12 Sample Size = 50
AC-0 Re-1

"

Page 23

Figure 2, top block, change "MK 50 Mod 1 or 2" to "MK 57 Mod 1 or 2"

Both above and below the "Transportation Vibration" block, add:

"
Radiographic Inspection 4.5.12 Sample Size = 12
AC-0 Re-1

"

Page 24

Figure 3, top block, change "MK 50 Mod 1 or 2" to "MK 57 Mod 1 or 2"

After the "Dimensional and Material Inspection" block add:

"
Radiographic Inspection 4.5.12 100% Inspection; reject defectives

"

In four blocks, change 100% Inspection" to "100% Inspection; reject defectives".

Page 25

Figure 4, top block, change "MK 50 Mod 1 or 2" to "MK 57 Mod 1 or 2"

Both above and below the "Transportation Vibration block" add:

"
Radiographic Inspection 4.5.12 Sample Size = 12
AC-0 Re-1

"

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6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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